STEP 1 — Create Notebook

```
# Lab14.4_NER_spaCy_HF_V.Sravya_2403A52403
```

STEP 2 — Install Required Libraries

```
!pip install spacy transformers torch pandas datasets --quiet
!python -m spacy download en_core_web_sm
```

```
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 89.3 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

STEP 3 — Import Libraries

```
# spaCy for traditional NLP pipeline
import spacy
from spacy import displacy

# Hugging Face transformers for transformer-based NER
from transformers import pipeline, AutoTokenizer, AutoModelForTokenClassification

# PyTorch backend (required for transformers)
import torch

# For storing results in tabular format
import pandas as pd
```

STEP 4 — Load spaCy Model

```
# Load pretrained English model
nlp = spacy.load("en_core_web_sm")
```

STEP 5 — Prepare Sample Text Data

```
sentences = [
    "Microsoft announced a new AI product in California on Monday.",
    "Virat Kohli scored a century against Australia in Mumbai.",
    "The United Nations held a climate summit in New York.",
    "Tesla CEO Elon Musk launched Starlink in India.",
    "Google acquired YouTube for $1.65 billion in 2006."
]
```

STEP 6 — Apply spaCy NER

```
print("----- spaCy NER Results -----\n")

for sentence in sentences:
    doc = nlp(sentence)
    print(f"Sentence: {sentence}")
    for ent in doc.ents:
        print(f"Entity: {ent.text} | Label: {ent.label_}")
    print("-"*50)
```

```
----- spaCy NER Results -----

Sentence: Microsoft announced a new AI product in California on Monday.
Entity: Microsoft | Label: ORG
Entity: AI | Label: ORG
Entity: California | Label: GPE
Entity: Monday | Label: DATE
--------------------------------------------------
Sentence: Virat Kohli scored a century against Australia in Mumbai.
Entity: Virat Kohli | Label: PERSON
Entity: a century | Label: DATE
Entity: Australia | Label: GPE
Entity: Mumbai | Label: GPE
```

```
--------------------------------------------------
Sentence: The United Nations held a climate summit in New York.
Entity: The United Nations | Label: ORG
Entity: New York | Label: GPE
--------------------------------------------------
Sentence: Tesla CEO Elon Musk launched Starlink in India.
Entity: Tesla | Label: ORG
Entity: Elon Musk | Label: PERSON
Entity: Starlink | Label: ORG
Entity: India | Label: GPE
--------------------------------------------------
Sentence: Google acquired YouTube for $1.65 billion in 2006.
Entity: Google | Label: ORG
Entity: YouTube | Label: ORG
Entity: $1.65 billion | Label: MONEY
Entity: 2006 | Label: DATE
--------------------------------------------------
```

STEP 7 — Store spaCy Results in DataFrame

```
spacy_results = []

for sentence in sentences:
    doc = nlp(sentence)
    for ent in doc.ents:
        spacy_results.append([sentence, ent.text, ent.label_])

spacy_df = pd.DataFrame(spacy_results, columns=["Sentence", "Entity", "Label"])

spacy_df
```

	Sentence	Entity	Label
0	Microsoft announced a new AI product in Califo...	Microsoft	ORG
1	Microsoft announced a new AI product in Califo...	AI	ORG
2	Microsoft announced a new AI product in Califo...	California	GPE
3	Microsoft announced a new AI product in Califo...	Monday	DATE
4	Virat Kohli scored a century against Australia...	Virat Kohli	PERSON
5	Virat Kohli scored a century against Australia...	a century	DATE
6	Virat Kohli scored a century against Australia...	Australia	GPE
7	Virat Kohli scored a century against Australia...	Mumbai	GPE
8	The United Nations held a climate summit in Ne...	The United Nations	ORG
9	The United Nations held a climate summit in Ne...	New York	GPE
10	Tesla CEO Elon Musk launched Starlink in India.	Tesla	ORG
11	Tesla CEO Elon Musk launched Starlink in India.	Elon Musk	PERSON
12	Tesla CEO Elon Musk launched Starlink in India.	Starlink	ORG
13	Tesla CEO Elon Musk launched Starlink in India.	India	GPE
14	Google acquired YouTube for $1.65 billion in 2...	Google	ORG
15	Google acquired YouTube for $1.65 billion in 2...	YouTube	ORG
16	Google acquired YouTube for $1.65 billion in 2...	$1.65 billion	MONEY
17	Google acquired YouTube for $1.65 billion in 2...	2006	DATE

Next steps: Generate code with spacy_df New interactive sheet

STEP 8 — (Optional) Visualize Entities

```
doc = nlp(sentences[0])
displacy.render(doc, style="ent", jupyter=True)
```

Microsoft ORG announced a new AI ORG product in California GPE on Monday DATE .

STEP 9 — Load Hugging Face NER Model

```
model_name = "dbmdz/bert-large-cased-finetuned-conll03-english"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForTokenClassification.from_pretrained(model_name)
```

```
ner_pipeline = pipeline("ner", model=model, tokenizer=tokenizer)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100%  998/998 [00:00<00:00, 31.0kB/s]
tokenizer_config.json: 100%  60.0/60.0 [00:00<00:00, 2.12kB/s]
vocab.txt: 213k/? [00:00<00:00, 2.58MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
model.safetensors: 100%  1.33G/1.33G [00:18<00:00, 51.9MB/s]
Loading weights: 100%  391/391 [00:01<00:00, 393.21it/s, Materializing param=classifier.weight]
BertForTokenClassification LOAD REPORT from: dbmdz/bert-large-cased-finetuned-conll03-english
Key                      | Status     |  |
-------------------------+------------+--+-
bert.pooler.dense.bias   | UNEXPECTED |  |
bert.pooler.dense.weight | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

STEP 10 — Apply Transformer NER

```
print("----- Hugging Face NER Results -----\n")

for sentence in sentences:
    results = ner_pipeline(sentence)
    print(f"Sentence: {sentence}")
    for r in results:
        print(f"Word: {r['word']} | Label: {r['entity']} | Score: {round(r['score'],3)}")
    print("-"*50)
```

```
----- Hugging Face NER Results -----

Sentence: Microsoft announced a new AI product in California on Monday.
Word: Microsoft | Label: I-ORG | Score: 0.9990000128746033
Word: AI | Label: I-ORG | Score: 0.890999972820282
Word: California | Label: I-LOC | Score: 1.0
--------------------------------------------------
Sentence: Virat Kohli scored a century against Australia in Mumbai.
Word: V | Label: I-PER | Score: 1.0
Word: ##ira | Label: I-PER | Score: 0.9959999918937683
Word: ##t | Label: I-PER | Score: 0.9990000128746033
Word: Ko | Label: I-PER | Score: 0.9990000128746033
Word: ##hl | Label: I-PER | Score: 0.9900000095367432
Word: ##i | Label: I-PER | Score: 0.9940000176429749
Word: Australia | Label: I-LOC | Score: 1.0
Word: Mumbai | Label: I-LOC | Score: 0.9990000128746033
--------------------------------------------------
Sentence: The United Nations held a climate summit in New York.
Word: United | Label: I-ORG | Score: 0.9990000128746033
Word: Nations | Label: I-ORG | Score: 0.9990000128746033
Word: New | Label: I-LOC | Score: 0.9990000128746033
Word: York | Label: I-LOC | Score: 0.9990000128746033
--------------------------------------------------
Sentence: Tesla CEO Elon Musk launched Starlink in India.
Word: Te | Label: I-ORG | Score: 0.9909999966621399
Word: ##sla | Label: I-ORG | Score: 0.9819999933242798
Word: El | Label: I-PER | Score: 1.0
Word: ##on | Label: I-PER | Score: 0.9990000128746033
Word: Mu | Label: I-PER | Score: 0.9990000128746033
Word: ##sk | Label: I-PER | Score: 0.9900000095367432
Word: Star | Label: I-MISC | Score: 0.6259999871253967
Word: ##link | Label: I-MISC | Score: 0.5989999771118164
Word: India | Label: I-LOC | Score: 1.0
--------------------------------------------------
Sentence: Google acquired YouTube for $1.65 billion in 2006.
Word: Google | Label: I-ORG | Score: 0.9990000128746033
Word: YouTube | Label: I-ORG | Score: 0.9980000257492065
--------------------------------------------------
```

STEP 11 — Clean Transformer Output

```
hf_results = []

for sentence in sentences:
```

```
    results = ner_pipeline(sentence)

    for r in results:
        word = r['word'].replace("##", "")   # Remove subword marks
        label = r['entity']
        score = round(r['score'], 3)

        hf_results.append([sentence, word, label, score])

hf_df = pd.DataFrame(hf_results,
                     columns=["Sentence", "Entity", "Label", "Confidence Score"])

hf_df
```

	Sentence	Entity	Label	Confidence Score
0	Microsoft announced a new AI product in Califo...	Microsoft	I-ORG	0.999
1	Microsoft announced a new AI product in Califo...	AI	I-ORG	0.891
2	Microsoft announced a new AI product in Califo...	California	I-LOC	1.000
3	Virat Kohli scored a century against Australia...	V	I-PER	1.000
4	Virat Kohli scored a century against Australia...	ira	I-PER	0.996
5	Virat Kohli scored a century against Australia...	t	I-PER	0.999
6	Virat Kohli scored a century against Australia...	Ko	I-PER	0.999
7	Virat Kohli scored a century against Australia...	hl	I-PER	0.990
8	Virat Kohli scored a century against Australia...	i	I-PER	0.994
9	Virat Kohli scored a century against Australia...	Australia	I-LOC	1.000
10	Virat Kohli scored a century against Australia...	Mumbai	I-LOC	0.999
11	The United Nations held a climate summit in Ne...	United	I-ORG	0.999
12	The United Nations held a climate summit in Ne...	Nations	I-ORG	0.999
13	The United Nations held a climate summit in Ne...	New	I-LOC	0.999
14	The United Nations held a climate summit in Ne...	York	I-LOC	0.999
15	Tesla CEO Elon Musk launched Starlink in India.	Te	I-ORG	0.991
16	Tesla CEO Elon Musk launched Starlink in India.	sla	I-ORG	0.982
17	Tesla CEO Elon Musk launched Starlink in India.	El	I-PER	1.000
18	Tesla CEO Elon Musk launched Starlink in India.	on	I-PER	0.999
19	Tesla CEO Elon Musk launched Starlink in India.	Mu	I-PER	0.999
20	Tesla CEO Elon Musk launched Starlink in India.	sk	I-PER	0.990
21	Tesla CEO Elon Musk launched Starlink in India.	Star	I-MISC	0.626
22	Tesla CEO Elon Musk launched Starlink in India.	link	I-MISC	0.599
23	Tesla CEO Elon Musk launched Starlink in India.	India	I-LOC	1.000
24	Google acquired YouTube for $1.65 billion in 2...	Google	I-ORG	0.999
25	Google acquired YouTube for $1.65 billion in 2...	YouTube	I-ORG	0.998

Next steps: Generate code with hf_df New interactive sheet

Start coding or generate with AI.

Double-click (or enter) to edit

Start coding or generate with AI.